



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri,
Nanded-431606.

Department of Computer Science & engineering.

Academic Year: 2025-26 (semester I)

End Semester Examination

Class: Final Year B.Tech.

Date: 27.11.2025

Course Name: Elective: Data Mining

Time: 13:30 to 15:30

Course Code: PEC-CS408

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1 a	Briefly describe and give examples of each of the following approaches to clustering : I. portioning methods . II. hierarchical methods . III. density-based methods .	06	CO1	BT2
b	Explain Bayesian classification method with appropriate example and briefly outline the major ideas of naïve Bayesian classification.	06	CO3	BT1
Q.2 a	Explain K-means clustering algorithm.	06	CO2	BT1
b	Consider the following database consisting of 6 transactions:	06	CO4	BT3

Sr.No	Transaction Id	Items
1.	101	A,C,D
2.	102	B,C,D,E
3.	103	B,D,E
4.	104	B,C,E
5.	105	A,B,D
6.	104	B,C,D

Suppose minimum support count is fixed as 50% and minimum confidence required is 70%. Find out the frequent item set using the Apriori algorithm. Also generate the associate rules for above given data.

Q.3 a Suppose a group of 12 *sales price* records has been sorted as follows:
5,10,11,13,15,35,50,55,72,92,204,215.

06

CO2

BT3

Partition them into three bins by each of the following methods:

(a) equal-frequency (equal-depth) partitioning

(b) equal-width partitioning

(c) clustering

b Suppose that the data mining task is to cluster points (with (x,y) representing location) into three clusters, where the points are
A1.(2, 10) , A2(2, 5) , A3(8, 4) , B1(5, 8) , B2(7, 5) , B3(6, 4) , C1(1, 2) ,
C2(4, 9) .

06

CO3

BT2

The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use the *k-means* algorithm to show *only*

(a) The three cluster centers after the first round of execution.

(b) The final three clusters.

Q.4 Explain Data Mining. Why is Data Mining said to be a multi-disciplinary field? Explain the architecture of a data mining system with a suitable diagram.

12

CO1

BT1

Q.5 Why is tree pruning useful in decision tree induction? What is a drawback of using a separate set of tuples to evaluate pruning?

12

CO5

BT4

Given a decision tree, you have the option of

(a) *converting* the decision tree to rules and then pruning the resulting rules, or

(b) *pruning* the decision tree and then converting the pruned tree to rules.

What advantages (a) will have over (b)?

ALL THE BEST



**Shri Guru Gobind Singhji Institute of Engineering
and Technology, Nanded-431606**

End Term Exam
Date : 22/11/2025

Subject: **Distributed Computing**
Time:13.30 To 15.30

Code: **PCC-CS404**
Max. Marks: 60

Instructions:

1. All Questions are compulsory.
2. Assume suitable data if necessary.
3. Figures to the right indicate full marks.

- Q.1 Write in brief to solve the Following.** 12
- a. Compare MapReduce with traditional RDBMS. CO1 BT1
 - b. What is file system model? Explain file system model for NFS. CO2 BT1
- Q.2** 12
- a. Why do persistent connections generally improve performance compared to nonpersistent connections? CO4 BT2
 - b. Explain MIME and its types. CO3 BT1
- Q.3** 12
- a. What is a Object Adapter? Explain Object Adapter in detail. CO2 BT2
 - b. What are the Enterprise Java Beans? Explain types of EJB. CO4 BT5
- Q.4 Write in brief to solve the Following.** 12
- a. What are the web services? Explain the principle of Web Service. CO2 BT1
 - b. Explain the cluster based architecture of DFS that is often used for parallel applications. CO5 BT2
- Q.5 Write Short Notes on (Any two)** 12
- a. The general organization of the Apache Web server. CO1 BT1
 - b. Resource record types in DNS. CO3 BT2
 - c. Hadoop Distributed File System CO2 BT3

End



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of computer science & engineering
Academic Year: 2025-26 (semester I)
End Semester Examination

Class: Final Year B.Tech.	Course Name: Cryptography and Network Security	Course Code: PCC-CS-403
Date: 20-11-2025	Time: 13:30-15:30	Max. Marks: 60

Note:

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- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		SOLVE ALL QUESTIONS	Marks	CO	BT
Q.1	a	Define Secret Key Cryptography and Public Key Cryptography.	04	CO2	BT2
	b	Test whether 97 is prime using any primality testing method.	04	CO1	BT3,4
	c	Explain Digital Signature and its working with a neat diagram.	04	CO4	BT2
	d	Explain the working of SSL/TLS in web security.	04	CO5	BT2
	e	Define Message Authentication Code (MAC). Mention its applications.	04	CO3	BT1
Q.2	a	Construct a Playfair matrix using the key "MONSOON" and encrypt "CLOUDY"	05	CO2	BT3
	b	Write notes on Firewalls and their types.	05	CO5	BT1
	c	Solve the system using CRT: $x \equiv 3 \pmod{5}$, $x \equiv 4 \pmod{7}$	05	CO1	BT3,4
	d	Explain the architecture and working of Kerberos.	05	CO4	BT1,2
Q.3	a	Given: - p=17, q=11 - Public exponent e=7 - Message M=8 Tasks: a) Calculate $\phi(n)$ and $\phi(n)$ b) Compute private key d c) Encrypt M d) Decrypt the ciphertext (Show complete steps.)	10	CO1	BT3,4
	b	Describe the Feistel cipher structure. Discuss how DES uses Feistel rounds. Why is the Feistel structure considered secure and flexible for symmetric key design?	10	CO2	BT1,2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the number theory concepts
CO 2	Impart knowledge on symmetric and asymmetric encryption techniques.
CO 3	Create Authentication functions as Message Authentication Codes and Hash Functions.
CO 4	Examine the issues and structure of Authentication Service and Electronic Mail, web and IP security
CO 5	Provide familiarity in system security.



Date: 25/11/2025

SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

End Semester Examination (Semester-I) 2025
Final Year B. Tech.
Computer Science & Engineering
TCP/IP Networking
PCC-CS402

Time: 13:30 to 15:30

Max. Marks : 60

- Note: i. Attempt all questions
ii. Assume suitable data if necessary
iii. Use of non-programmable calculator is permitted.

Q.N.	Question	Marks	CO	BT
Q.1	What are the different classes of IP addresses. Give the range of addresses for each class.	8	CO1	BT4
Q.2	Define and explain Classless addressing with suitable example.	6	CO2	BT1
Q.3	What is Subnetting? How the subnets are designed? Give suitable example.	8	CO3	BT6
Q.4	Compare Direct Versus Indirect Delivery. Give suitable example for each case.	6	CO4	BT4
Q.5	Show the IP Datagram format and explain the functions of all fields.	8	CO5	BT2
Q.6	How does an ICMP gives report about destination unreachable error? Explain with suitable ICMP format and example.	8	CO5	BT1
Q.7	Demonstrate the UDP Checksum calculation with suitable example.	8	CO6	BT2
Q.8	Explain the working of TCP and its connection establishment process with suitable example.	8	CO6	BT5

Course Outcomes:

- CO 1: Understand the TCP IP protocol suite and IP addressing.
CO 2: Know the concepts of Classful and Classless IP addressing.
CO 3: Design the subnets.
CO 4: Learn IP Packet forwarding and routing table.
CO 5: Study the importance of different Network layer protocols (ARP, IP, ICMP etc..)
CO 6: Learn the use of different Transport and Application level Protocols(UDP, TCP, RIP, DHCP, HTTP etc..)

Blooms Taxonomy:

- BT1: Remember, BT2: Understand, BT3: Apply, BT4: Analyze, BT5: Evaluate, BT6: Create